POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Executive Summary of the Thesis

Entropy: Uniting Statistichal Mechanics and Thermodynamics

Laurea Magistrale in Aeronautical Engineering - Ingegneria Industriale e dell'Informazione

Author: Michele Andreozzi - 10665771

Advisor: Prof. Giulio Gori

Academic year: 2023-2024

---

1. Introduction

This short summary is intended to provide MSc students with an intuitive, yet formal, description of the details that govern the definition and study of entropy.
Oftentimes entropy is presented as a "measure of disorder", leaving doubts whether such a vague concept can be captured by the precision of mathematics. The goal is therefore that of developing a description that naturally leads to both microscopic and macroscopic definitions of entropy and its properties.
The summary is developed as follows: in section 2 the various definitions of entropy, given in different contests of science, are presented and connections between them are highlighted. In section 3 the aforementioned, deterministic, descriptions of entropy are tied to the second law of thermodynamics thanks to the H theorem. A modern appraisal of this theorem is developed, so that criticisms that emerged through the years can be fully appreciated. Finally, in section 4, a quantitative analysis, touching on all concepts presented throughout, is presented by means of a fully deterministic simulation code.

2. What is entropy?

2.1. In Information Theory

In his 1948 breakthrough work "A Mathematical Theory of Communication" [14] Claude Shannon, building on the attempts of Nyquist and Hartley [9, 12], formalized a theory of information. A few original statements from Shannon can enlighten the reader as to what the purpose of such formalization was:

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. [...] The significant aspect is that the actual message is one selected from a set of possible messages. [...] If the number of messages in the set is finite then this number or any monotonic function of this number can be regarded as a measure of the information produced when one message is chosen from the set, all choices being equally likely.

It is therefore a question of finding how much knowledge is gained in the act of receiving and deciphering a message. The proposal coming from Shannon is that of identifying the increase in knowledge with the decrease in uncertainty; such a formulation naturally leads to the introduction of a theory in terms of probabilities,

where each statement can be characterized by a number representing the confidence in the statement being true. A simple example from Blundell [2] will certainly clarify:

- Isaac Newton's birthday falls on a particular day of the year.
- Isaac Newton's birthday falls in the second half of the year.
- Isaac Newton's birthday falls on the 25th of a month.

The three statements respectively have a $\frac{1}{1}$, $\frac{1}{2}$ and $\frac{12}{365}$ probability ob being true. The key observation is that the statement with the least probability reduces the number of possible correct answers the most, it is therefore the message containing the most information. A further property required by information is that the information carried by two statements will be the sum of the two information contents. Notice that the probability of two statements being correct at the same time follows the rules of multiplication [1] (e.g. the probability of Isaac Newton's birthday falling on the 25th of a month in the second half of the year is $\frac{1}{2}\frac{12}{365} = \frac{6}{365}$). The function that transforms products into sums in obviously the logarithm, and thus Shannon suggested the following definition for the information content:

$$Q_i = -k \log p_i \tag{1}$$

Where Q_i is the information contained in the i-th message and p_i is the probability of that same statement.[2]. The choice of the basis comes from practicality only: in communication using $\log_2$ allows to express the content in terms of bits of information, while the natural logarithm is the best option when dealing with natural sciences. The proportionality constant k can be adapted for each scenario, and in Shannon's theory it assumes unitary value.
Once a set of statements is collected one can then define "Shannon's entropy" as the average information content of the set, i.e.:

$$S = \sum_i p_i Q_i = -k \sum_i p_i \log p_i \tag{2}$$

[1] Under the hypothesis of uncorrelated probabilities for the two messages.

[2] Notice the minus sign is there since p_i cannot exceed the value of 1, thus becoming negative under the logarithm. In this way the least probable message has the largest, positive, information content.

In this way Shannon's entropy represents the amount of uncertainty in a system. A set where probabilities are equally spread out consequently has more entropy than a set with a single highly probable outcome: the result is more predictable in the second case, meaning it has less uncertainty.

2.2. In Statistical Mechanics

From Peliti's work "Statistical Mechanics in a nutshell" [13]:

The goal of statistical mechanics is to predict the macroscopic properties of bodies, most especially their thermodynamic properties, on the basis of their microscopic structure.

A cornerstone in the whole theory of statistical mechanics is the alternating use of the contrasting concepts of microstate and macrostate. Following [2] we will try to build these notions starting from the idea of thermal equilibrium, aiding ourselves with two thought experiments going hand in hand.
For the first experiment we take two isolated bodies and put them in contact. Experience tells us that this will result in the process called "thermalization", i.e. an heat flow is generated until a state of equilibrium is reached.
Let us assume that the two bodies are made of a large number of smaller, more fundamental objects, and that each object can occupy a state from a set of possible ones. The number of possible configurations for the ensemble of smaller objects is exponentially larger than the number of objects themselves. If we further assume, however, that these fundamental building blocks are all alike, then a very large amount of possible configurations will be indistinguishable from one another. Moreover there will also be many configurations which will be immensely similar.
We now say that every single configuration is a microstate of the system, whereas a set of indistinguishable microstates is called a macrostate. Notice that for each macrostate we can, with the aid of both a model for the system at hand and of statistics, enumerate the number of possible indistinguishable microstates. This number will be denoted by Ω.

In the example of the two bodies in contact

we say that, since the heat involved in thermalization is a form of energy flow, the observable macrostate is that of energy, denoted by E. We previously formulated that the number of microstates is a function of the corresponding macrostate, therefore we have $\Omega = \Omega(E)$, and so for our two bodies:

$$\Omega_i(E_i) \qquad \forall i = 1, 2 \tag{3}$$

If these two bodies can only exchange energy, and keeping the system as a whole isolated, we can say that:

- the total amount of energy will be conserved, i.e. $E = E_1 + E_2 = \text{const}$
- equilibrium will be reached as soon as $\frac{\partial E_1}{\partial t} = \frac{\partial E_2}{\partial t} = 0$

Probability theory and ergodic theory, specifically Poincaré recurrence theorem [5], tell us that in time the system will explore all possible, equally likely, microstates. Notice that many of these explored microstates will be indistinguishable from one another, thus forming different macrostates.
The last statements may seem in contradiction: we said equilibrium is reached when a final, non varying, macrostate is reached, while on the other hand we said all microstates, even those with a different macrostate, will come up in time.
It is now that probability and information theory can help. We will thus conceive a second though experiment, that will hopefully enlighten the reader on the orders of magnitude involved in the present discussion. Suppose we have a memory of 100 bits[3], each with equally likely possible states 1 and 0. We then have 2^{100} possible microstates for the memory. The measurable macrostate is chosen to be $\mathcal{N}$: the number of bits taking value 1. From probability:

$$\Omega(\mathcal{N}) = \frac{100!}{\mathcal{N}!(100-\mathcal{N})!} \tag{4}$$

The behaviour of this discrete function with $\mathcal{N}$ is plotted in figure 1.

[3]Notice that physical systems tend to have dimensions in the order of at least $\mathcal{O}(10^{23})$ microscopic objects, immensely more than 100.

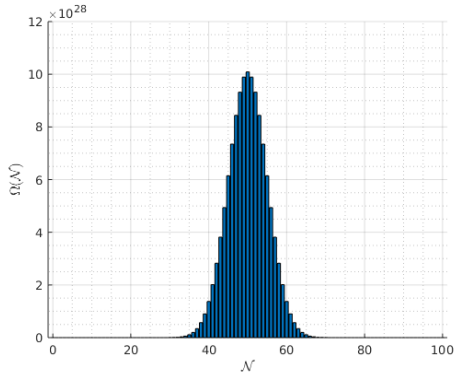

Figure 1: The number of microstates as function of the macrostate

A few noteworthy observations are:

- there is a single microstate for the macrostate $\mathcal{N} = 100$, i.e. $\Omega(100) = 1$. The same goes for the macrostate $\mathcal{N} = 0$
- the maximum number of microstates corresponds to the macrostate $\mathcal{N} = 50$, for which we we have $\Omega(50) \approx 1 \times 10^{29}$
- it is approximately 3500 times more likely to find the macrostate $\mathcal{N} = 50$ then the macrostate $\mathcal{N} = 30$
- the width of the Gaussian "bell curve" decreases as the number of objects increases from 100, thus it is less and less likely to find the system in a macrostate different from the most probable one, even the ones just adjacent to it

Defining:

$$\mathcal{N}_{max} = \mathcal{N} \quad s.t. \quad \Omega(\mathcal{N}_{max}) = \Omega_{max} \tag{5}$$

In Vincenti and Kruger book "Introduction to Physical Gas Dynamics" [16] it is shown that $\frac{\Omega(\mathcal{N}_{max\pm1})}{\Omega_{max}} < e^{10^{-13}}$ for the isolated system composed by 1 cm^3 of air at 273.15 K and 101325 Pa. The dimensions and complexity of such a system pale in comparison with usual systems found in engineering problems. This should definitively convince any reader that, once equilibrium is reached, it is practically impossible to find the system in any macrostate different from the most likely one. Further, if, following the reasoning of [16], one makes the physical assumption that the hypothesis of equal probability for the different microstates corresponds to an equal amount of time spent in each microstate while exploring the full phase space, it is then obvious

to deduce that the system must spend the vast majority[4] of time in the macrostate $\mathcal{N}_{max}$. It is therefore natural, remembering the definition of approach to equilibrium given at page 3, to say that the equilibrium macrostate is the one characterized by the largest number of corresponding microstates.

We can now go back to the first thought experiment, where the thermalization process of two bodies was studied. In order to find the equilibrium state we will need to maximize the total number of microstates, i.e. $\Omega = \Omega_1(E_1)\Omega_2(E_2)$, thus:

$$\frac{d\Omega_{max}}{dE_1} = \frac{d}{dE_1}[\Omega_1(E_1)\Omega_2(E_2)] = 0 \tag{6}$$

Now thanks to energy conservation we can write $E_2 = E_2(E_1) = E - E_1$, so that:

$$\frac{d\Omega_{max}}{dE_1} = \Omega_1 \frac{d\Omega_2}{dE_2}\frac{dE_2}{dE_1} + \Omega_2 \frac{d\Omega_2}{dE_1} = 0 \tag{7}$$

And since $\frac{dE_2}{dE_1} = -1$ we easily get:

$$\frac{1}{\Omega_1}\frac{d\Omega_1}{dE_1} = \frac{1}{\Omega_2}\frac{d\Omega_2}{dE_2} \tag{8}$$

Integrating we can find the condition for equilibrium in statistical mechanics terms:

$$\frac{d}{dE_1}\log\Omega_1 = \frac{d}{dE_2}\log\Omega_2 \tag{9}$$

We now remember that in everyday experience equilibrium is characterized by the bodies reaching the same temperature [5], so it seems obvious to rewrite this equilibrium condition as an equality of temperatures. It turns out, for reasons that will soon be clear, that it is convenient to use equation (9) to define the reciprocal of temperature:

$$\frac{1}{T} := k_B \frac{d(\log\Omega)}{dE} \tag{10}$$

Where the constant k_B, called Boltzmann constant, is used only so that the newly defined quantity has units of temperature, having value $k_B = 1.3807 \times 10^{-23} JK^{-1}$. Again using equation (9), we can also find:

$$dE = Td(k_B \log\Omega) \tag{11}$$

[4] As seen just above, such "vast majority" will, in the limit, become the complete totality of time for realistic systems.

[5] Actually this is often how equilibrium is presented in undergraduate level courses.

If we now allow ourselves to take a leap in the direction of classical thermodynamics we know that $dE = TdS - pdV$. But since we specified our two systems can only exchange "thermal" energy we have that $dV = 0$, and thus:

$$S = k_B \log\Omega \tag{12}$$

Relation (12) is certainly one of the most famous equations in physics history, and characterizes what is known as Boltzmann's entropy, defined for systems in thermodynamic equilibrium. Notice that, from equation (12), the higher the number of available microstates for a system, the higher it is its entropy. There is, however, a possible source of confusion, specifically in what the aforementioned Ω value in equation (12) should represent: it is the total number of microstates the system can have considering all possible macrostates or is it only the number of microstates associated with the measured macrostate?

Trying to clarify this will lead us to a second definition of entropy. Let us suppose we know:

- the number of microstates in each possible macrostate, denoted by m_i for the i-th macrostate
- the total number of possible microstates, denoted by $M = \sum_i m_i$

We then know that the probability of finding the system in the i-th microstate is given by:

$$p_i = \frac{m_i}{M} \tag{13}$$

We then say the "total entropy" of the system is $\mathfrak{S} = k_B \log M$, yet only a part of this can be measured since we are not able to distinguish indiviual microstates[6], but only macrostates characterized by different macroscopic properties:

$$\mathfrak{S} = S + S_{micro} \tag{14}$$

Where S_{micro} is the part of "total entropy" that we cannot measure. What we cannot measure

[6] It is not our interest either, as we are only able to perceive macroscopic properties of matter. Paradoxically the "total entropy" of the whole universe, taken as an isolated system, would not change in time, as the total number of available microstates considering all possible macrostates is always the same. What changes is the macrostate we find the universe in, and thus the measurable entropy

is that the system can be in any of the different microstates associated with a particular macrostate [2], and so we say:

$$S_{micro} = \sum_i p_i k_B \log m_i \tag{15}$$

Putting together equations (14) and (15) yields:

$$S = k_B(\log M - \sum_i p_i \log m_i) \tag{16}$$

Knowing $\sum_i p_i = 1$ it is found:

$$S = k_B \sum_i p_i(\log M - \log m_i) \tag{17}$$

Then using the properties of logarithms:

$$S = k_B \sum_i p_i \log \frac{M}{m_i} = -k_B \sum_i p_i \log \frac{m_i}{M} \tag{18}$$

Finally, since $\frac{m_i}{M} = p_i$, we express entropy as:

$$S = -k_B \sum_i p_i \log p_i \tag{19}$$

This is known as Gibbs entropy, and it can be shown [10] to be valid also in non equilibrium conditions, a property that will be used in section 3. It must now be pointed out that Gibbs' definition of entropy should immediately recall Shannon's entropy in information theory. On the basis of that development presented in subsection 2.1 it should feel natural to connect the idea of entropy with the idea of uncertainty, and thus it can be understood why entropy has long been associated with the idea of disorder, intended as lack of predictability. Notice finally that assuming $p_i = \frac{1}{\Omega}$ in (19) one immediately finds Boltzmann entropy, showing that at equilibrium the two definitions are equivalent.

Statistical mechanics is an extremely successful theory and very detailed accounts of its most important successes can be found in [5, 16] or in the countless monographs published through the years, yet one major achievements must be briefly shown.
We know that the perfect gas model describes a set of n non interacting particles. As shown in [5] the application of Boltzmann entropy formula to this case leads, through some omitted cumbersome manipulations, to:

$$S(n, V, E) = k_B n \left(\log \frac{V}{V_0} + \frac{3}{2} \log \frac{E}{E_0} \right) + C \tag{20}$$

Where V_0, E_0 and C are simple constants. In the next subsection this result in particular will prove that statistical mechanics yields an excellent microscopic description of reality, correctly predicting macroscopic properties of interest.

2.3. In Thermodynamics

In classical thermodynamics we say:

$$dS = \frac{đQ_{rev}}{T} \tag{21}$$

The notation $đQ$ indicates that heat is an imperfect differential, yet it becomes a state function when computed along a reversible transformation, thus entropy is defined as a state function. Recalling Clasius inequality [6] one easily gets:

$$dS \geq \frac{đQ}{T} \tag{22}$$

So that in isolated systems $dS \geq 0$, a mathematical statement of the second law of thermodynamics.
In 1970 Galgani and Scotti, through the paper "On subadditivity and convexity properties of thermodynamic functions" [7] unified thermodynamics under a very simple and axiomatic framework. In their treating of the subject it is shown that the whole thermodynamic description of a system can be deduced from one of two quantities, identified as "thermodynamic potentials":

- the energy E of the system, $E = E(n, V, S)$
- the entropy S of the system, $S = S(n, V, E)$

Properties of strict monotonicity in either S or E allow to invert the functional relation, while the property of superadditivity (subadditivity) of entropy (energy) enforce the second law of thermodynamics. Once the functional relation $S(n, V, E)$ is known, all thermodynamic properties are easily derived. It should not be surprising, then, that equation (20) is exactly this functional relation, thanks to which thermodynamics and statistical mechanics are fully unified.

3. The H-Theorem and the Second Law of Thermodynamics

3.1. A thorough proof of the H theorem

The final theoretical necessity is formal proof of Boltzmann's famous H-theorem, for which we will follow Cercignani's work [4] for its clarity in conciseness. The theorem is built upon the notion of the "gas distribution function" f, so that in phase space $dN = f d\vec{r} d\vec{v}$. We define:

$$\mathcal{H} = \int_{\mathcal{D}_v} f \log f d\vec{v} \tag{23}$$

From section 2 it should be clear that this definition is related to Gibbs entropy formula by simply transforming the sum into an integral[7], as p_i is a discrete probability, while f is a continuous distribution. The H-theorem investigates the problem of non equilibrium thermodynamics, and the approach to equilibrium, thus an evolution equation for f is required. We of course know that this equation is precisely Boltzmann equation:

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla_{\vec{r}} f = \mathcal{C}(f, f) \tag{24}$$

If multiply the equation by $\log f$ and integrate in $d\vec{v}$ we then obtain:

$$\frac{\partial \mathcal{H}}{\partial t} + \nabla_{\vec{r}} \cdot \vec{\mathcal{J}} = \mathcal{I} \tag{25}$$

Where $\vec{\mathcal{J}}$ and $\mathcal{R}$ represent, respectively:

- $\vec{\mathcal{J}} = \int_{\mathcal{D}_v} \vec{v} f \log f d\vec{v}$ is the flux of $\mathcal{H}$
- $\mathcal{I} = \int_{\mathcal{D}_v} \mathcal{C}(f, f) \log f d\vec{v}$ is the production[8] of $\mathcal{H}$ due to the collision integral $\mathcal{C}(f, f)$

We then apply the symmetries of the collision integral, specifically the ones with respect to:

- quantities computed before and after the collision
- the exchange of the bullet particle with the target particle

So that one can then find:

$$\begin{aligned} &\int_{\mathcal{D}} \mathcal{C}(f, f) \log f d\vec{v} = \\ &\frac{1}{4} \int \int \int_{\mathcal{D}_v} (f_+ f'_+ - f f') \log(\frac{f_+ f'_+}{f f'}) \dots \\ &\dots \frac{||\vec{p_r}||}{m} \sigma(\vec{p_r}, \vartheta, \varepsilon) d\vec{v} d\vec{v}' d\vartheta d\varepsilon \end{aligned} \tag{26}$$

[7] Not considering, of course, the dimensional constant k_B and the minus sign. Actually $s_B = -k_B \mathcal{H}$ represents entropy per unit volume, from definition of f

[8] If negative it would represent a dissipation

Notice that $\forall y, z \in \mathbb{R}^+$ the elementary inequality $(z - y) \log(\frac{y}{z}) \leq 0$ is valid, so that:

$$\int_{\mathcal{D}_v} \mathcal{C}(f, f) \log f d\vec{v} \leq 0 \tag{27}$$

Which is known as Boltzmann inequality. It can also easily be shown that $\mathcal{I} = 0$ if and only if $\log f$ is a collision invariant, i.e. $\log f = a + \vec{b} \cdot \vec{v} + c||\vec{v}||^2$, where $a, \vec{b}, c$ are generic scalars. It is trivial then to derive that the only gas distribution function that guarantees $\mathcal{I} = 0$ is the Maxwellian $M = Ae^{-\beta||\vec{v}-\vec{u}||^2}$.

At this point it is convenient to introduce the quantity $H = \int_{\mathcal{D}_r} \mathcal{H} d\vec{r}$. Notice that in this case $S_B = -k_B H$ can be seen to represent the total amount of entropy in the region $\mathcal{D}_r$ of space. Equation (25) becomes then:

$$\frac{dH}{dt} \leq \int_{\partial \mathcal{D}_r} \vec{\mathcal{J}} \cdot \vec{n} d\omega \tag{28}$$

Where $d\omega$ is an infinitesimal element of the boundary $\partial \mathcal{D}_r$. Under periodic or reflection boundary conditions on $\partial \mathcal{D}_r$, that define an isolated system as discussed in detail in [4], the right hand side of equation (29) goes to zero, thus proving:

$$\frac{dH}{dt} \leq 0 \tag{29}$$

One last development is needed to formally prove the H theorem: let us study the difference $\mathcal{H} - \mathcal{H}_M = \int_{\mathcal{D}_v} f \log f - M \log M d\vec{v}$. If we add and subtract both $f \log M$ inside the integral we get:

$$\begin{aligned} \mathcal{H} - \mathcal{H}_M = &\int_{\mathcal{D}_v} (f - M) \log M d\vec{v} + \dots \\ &\dots + \int_{\mathcal{D}_v} f \log f - f \log M d\vec{v} \end{aligned} \tag{30}$$

Since $\log M$ is a collision invariant the first integral vanishes. We now add and subtract $M - f$, and apply the inequality $z(\log z - \log y) + y - z \geq 0$, valid for all real nonnegative y, z, so that:

$$\begin{aligned} &\int_{\mathcal{D}_v} f(\log f - \log M) + M - f d\vec{v} + \dots \\ &\dots + \int_{\mathcal{D}_v} f - M d\vec{v} \geq \int_{\mathcal{D}_v} f - M d\vec{v} = \dots \\ &\dots = n - n = 0 \end{aligned} \tag{31}$$

The last passage used the fundamental property $\int_{\mathcal{D}_v} f d\vec{v} = n$, valid for any distribution, thus also

for the Maxwellian M. This results in:

$$\mathcal{H} \geq \mathcal{H}_M \tag{32}$$

With this result we have found that $\mathcal{H}$, and consequently its spacial integral H, is a non increasing function of time, bounded from below by the equilibrium value $\mathcal{H}_M$. Under appropriate regularity conditions for the source term $\mathcal{I}$ that appears in 25, thoroughly analyzed in [1], we can then conclude that:

$$\lim_{t\to\infty} \mathcal{H} = \mathcal{H}_M \tag{33}$$

In terms of entropy the converse is true, so that any isolated system evolves in the direction of non decreasing entropy, until a final state of equilibrium is reached and the Maxweillian distribution homogeneously characterizes the system. This concludes a complete formal proof of Boltzmann H theorem, so that its critical aspects can be discussed.

3.2. Loschmidt's Paradox Discussion

The main source of critisicm to Boltzmann H theorem came from Jan Johann Joseph Loschmidt, who argued that the theorem proved a non time reversal invariant behaviour, yet coming from the time reversible dynamical laws of Newton. If the system is evolved for a certain amount of time, and then all particle velocities are reversed then the system would necessarily go back to its initial state, a behaviour allowed by dynamics, that apparently would cause entropy to revert back. It can be seen, as in [15], that the source of such behaviour is due to the *stosszahlansatz* assumption in the Boltzmann equation derivation. Such hypothesis is needed for particles just about to collide, while it is not assumed for particles that have just collided. Notably, if the opposite were true then the H theorem would yield an opposite result, meaning a non increasing entropy.
The *stosszahlansatz*, at its core, assumes that:

$$f^{(2)}(\vec{r_1}, \vec{v_1}, \vec{r_2}, \vec{v_2}) = f(\vec{r_1}, \vec{v_1}) f(\vec{r_2}, \vec{v_2}) \tag{34}$$

For particles just before the collision, thus simplifying the loss term in the collision integral. Thanks to the symmetry of the laws governing collisions we can however think of an inverse collision, and apply the *stosszahlansatz* to it [4]. The dispute between Boltzmann and Loschmidt then became purely probabilistic: how reasonable is this assumption that brings asymmetry with respect to time? Although very formal results have been obtained without any symmetry violation [11] they have very limited applicability. The paradox is purely a discussion on probable and improbable distributions, which may render the absence of correlation, used in equation (34), more or less plausible. As such a numerical assessment is performed in the following section.

4. Numerical Simulations

The coming simulations were intended to evaluate the quality of probability arguments in describing the dynamical evolution, therefore the code needed to be fully deterministic. As a consequence a collision simulator code, details of which are found in appendix A, has been developed for the 2D case of a closed box containing a system of hard spheres. An image representing the situation can be found in image 2. In order to reduce the computational cost, the code follows [8], computing the time of the first ensuing collision and advancing the explicit laws of motion of the particles by this exact time step. The initial conditions were randomly generated both in position and velocity, from an homogeneous distribution, i.e. a non Maxwellian, around a selectable reference velocity.

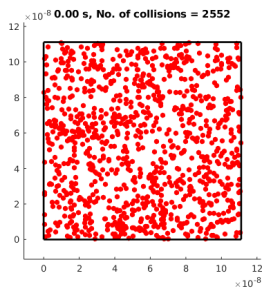

Figure 2: An image taken from the collision simulator in runtime

The first test intended to show the validity of Loschmidt's argument: 1000 particles were evolved for 100,000 collisions (much longer than the expected relaxation time $T_{relax} \approx \frac{5}{\pi d^2 n <||\vec{c}||>}$), then their velocities were reversed and another 100,000 collisions were simulated. In order to reproduce a realistic diluted gas the particle diameter was set to $d = 4\times10^{-10}$ m, the

length of the box to $L = 1 \times 10^{-7}$ m, and a reference velocity of $V_{ref} = 200$ m/s was chosen. A discrete velocity distribution function was computed, and used to evaluate the results. As expected, starting from a non Maxwellian f, the system evolved towards the Maxwell-Boltzmann distribution seen in figure 3. Thanks to the time symmetry of the Newton laws it is known that reversing all the particle velocities must result in the in the same exact non Maxwellian distribution that characterized the initial conditions. It should feel surprinsing that the simulation did not yield this result for this simulation. Figure 4 indeed shows that starting from a Maxwell-Boltzmann distribution (notice this is the final condition of the previous simulation), the system remains at the equilibium PDF.

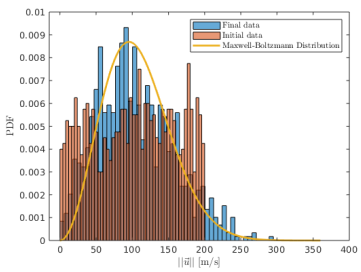

Figure 3: The initial distribution, in red, and the Maxwellian found at equilibrium, in blue

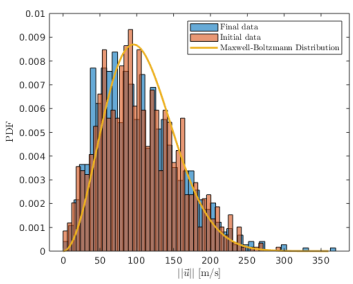

Figure 4: In red, the final condition of the previous test. In blue, the equilibrium condition found when reversing time

By repeatedly testing the code it was found that prefect time reversal could be obtained only with a very small amount of particles. Moreover this reversal was not a systematic result each time the inverse initial conditions were applied: further research confirmed that when particles are very close, to each other or to a wall, the computation of the necessary time step truncates numbers that are extremely small, and thus little errors in the time advancement propagate. This deviation is a remarkable result, showing that the smallest changes to the initial condition, namely the ones caused by the floating point arithmetic when collisions are computed, cause large departures from the "time reversed" expected result. It is striking then, that the small deviations caused by the computations always caused an evolution towards the Maxwellian. Further, not only the state that guarantees reversal must be recreated with tremendously high accuracy (as just shown), it is also extremely unlikely to randomly obtain a distribution that even vaguely correlates to such "perfect reversal state".
This can be seen as a strong reinforcement to Boltzmann's discussion: it is true that specific states that violate the H-theorem exist, and they are not forbidden by the laws of dynamics, but even very minor variations to these states lead to an evolution that follows the theorem. It proves that the *stosszahlansatz* is at the very least extremely effective at capturing the cause of irreversible processes, and that violations to such hypothesis can be considered, in all practical applications, essentially impossible.

5. Conclusions

This work tried to show, with the least amount of simplifications that could still guarantee formally sound results:

- how information theory adopted the concept of entropy from thermodynamics, and why its definition of it helps in clarifying the connection of entropy with disorder.
- the powerful tool that statistical mechanics proved to be in the reconciliation of a microscopic description with the macroscopic one, given by thermodynamics.
- a formal proof to the H theorem for a non space-homogeneous solution, also highlighting the role of the *stosszahlansatz* in the emergence of irreversibility from a time symmetric set of dynamic equations.
- a numerical justification for the validity of the *stosszahlansatz* itself. This hopefully showed that, although theoretical proofs of its validity seem distant from the present state of kinetic theory [3], the intuition that guided Boltzmann's argument yields excellent practical results.

References

[1] L. Arkeryd. On the boltzmann equation part ii: The full initial value problem. *Archive for Rational Mechanics and Analysis*, 1972.

[2] S. Blundell, S.J. Blundell, and K.M. Blundell. *Concepts in Thermal Physics*. Comprehensive Assessment of Water Management in Agriculture. Oxford University Press, 2006.

[3] Harvey R. Brown and Wayne Myrvold. Boltzmann's h-theorem, its limitations, and the birth of (fully) statistical mechanics, 2008.

[4] C. Cercignani. *Rarefied Gas Dynamics, From Basic Concepts to Actual Calculations*. Cambridge Texts in Applied Mathematics. Cambridge, 2012.

[5] A. Fasano and S. Marmi. *Meccanica analitica*. Programma di mat. fisica elettronica. Bollati Boringhieri, 2002.

[6] E. Fermi. *Termodinamica*. Programma di matematica, fisica, elettronica. Boringhieri, 1982.

[7] L. Galgani and A. Scotti. On subadditivity and convexity properties of thermodynamic functions. *Pure and Applied Chemistry*, 22(3-4):229–236, 1970.

[8] J.M. Haile. *Molecular Dynamics Simulation: Elementary Methods*. Wiley, 1997.

[9] R. V. L. Hartley. Transmission of information. *Bell System Technical Journal*, 7:535–563, 1928.

[10] E. T. Jaynes. Gibbs vs boltzmann entropies. *American Journal of Physics*, 33:391–398, 1965.

[11] O.E. Lanford. Time evolution of large classical systems. *Dynamical Systems, Theory and Applications. Lecture Notes in Theoretical Physics 38*, pages 1–111, 1975.

[12] Harry Nyquist. Certain factors affecting telegraph speed. *Journal of the A.I.E.E.*, 43:124–130, 1924.

[13] L. Peliti. *Statistical Mechanics in a Nutshell*. In a Nutshell. Princeton University Press, 2011.

[14] Claude Elwood Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27:379–423, 1948.

[15] Jos Uffink and Giovanni Valente. Lanford's theorem and the emergence of irreversibility. *Foundations of Physics*, 45(4):404–438, 2015.

[16] W.G. Vincenti and C.H. Kruger. *Introduction to Physical Gas Dynamics*. Wiley, 1965.

A. Appendix

The full code can be found at https://polimi365-my.sharepoint.com/:f:/r/personal/10665771_polimi_it/Documents/Fundamentals%20of%20Hypersonic%20Flows%20-%20Andreozzi?csf=1&web=1&e=Jlwse7, along with the final condition of the simulation presented in this work. The code can thus be used to both run a forward and a backward simulation.